Markets as the theory of rational action

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"Rationality" for a purely predictive agent (i.e. a **non-embedded** agent, *not* Predict-o-matic) is almost completely specified: Bayes's theorem.

But for agents that take actions (equivalently, those whose beliefs influence the world), the "utility function" is arbitrary. In active inference, this utility function is a "prior" e^U/Z (the agent's sense of identity or Self-model) — but unlike a usual Bayesian prior, this prior actually impacts the world and is therefore *hyperstitional*: it does not get dominated by evidence.

Well, utility/identity must ultimately be shaped by reward (e.g. RL or evolution). But reward, too, is "arbitrary".

Intuition for what a rational theory of reward should look like:

- "the universe as a whole" (or "closed systems") do not have identities, but individual sub-agents do
- identities are the result of the universe hierarchically enforcing its will on its sub-agents via reward/selection (though the question of exactly how actions, i.e. things causally determined by agent beliefs, emerge, should be clarified)
- reward has to do with constraints in statistical mechanics

My research can be summarized into three slogans:

1. **Reward is hyperstitional information.**
2. **Utility functions are a (path-dependent) sum of rewards.**
3. **Price recursion (backprop/Bellman) is the rational theory of reward and action.**

As a bonus: we demonstrate the long-sought market–neural networks/backprop equivalence and a market–MDP equivalence.

Future questions

I. Our markets still have utility functions at the end of the tunnel. **What is the "true" super-agent? Thermodynamics (free entropies) seems relevant to this.**

II. Markets are just a model, right? To prove that it is *the* rational theory of reward we'd need a **Dutch book type result for credit assignment**.

III. **Wealths and perfect competition.** A proper justification of continuous Bayesian inference as the dynamics of markets would require a game-theoretic study of Aumann-style infinitesimal agents maximizing local objectives and having wealths/budgets.

IV. **Dynamic rationality.** I want a rational theory of dynamics.

Reward is hyperstitional information

The most natural algorithm for learning in probability space is "entropic mirror ascent", i.e. gradient ascent but taking KL-divergence (equivalently the Fisher metric) instead of Euclidean distance as the "distance function". Where $f(p)$ is the objective (of policy p) we're trying to optimize, this looks like:

$$p_{t+1} \propto p_t \exp\big(\eta \nabla f(p_t)\big)$$

Or in differential form:

$$dp/dt = p(t) \odot \big(\nabla f(p) - p \cdot \nabla f(p)\big)$$

Optimizing some objective $f(p)$ over probability space with entropic mirror ascent, is continuous-time Bayesian inference with infinitesimal evidence $\exp(\eta \nabla f(p_t))$.

Unlike usual Bayesian inference:

1. this likelihood function depends on the current p
2. this likelihood function can be ... anything.

We know from active inference: "utility functions"/goals are actually the agent's prior: and **inference arbitrages between the agent's beliefs and the environment** (when the environment's information updates the agent that is "learning"; when the agent's information influences the environment that is "action").

And now we know reward too, is information. About what? About "what will I do". You can specify *any* objective whatsoever, and optimizing it (at least through entropic mirror ascent) will still be rational Bayesian inference. In other words, reward is *hyperstitional* information.

And if you multiply all the exponentiated rewards the agent gets through its training, you get its final prior, which expresses its exponentiated utility function.

In bandits, the objective is just the reward; in more general RL you must use Q-values, and also do importance sampling (i.e. use $f(p)/p_i$ where i is the taken action) because you only have access to the reward for one performed action. PPO methods used in modern RL are also approximately entropic mirror ascent (TRPO is exactly it). You may also recall that evolution (replicator dynamics) is Bayesian inference (Harper 2009) — indeed, it is exactly equivalent to this dynamic, with the fitness function as an exponentiated reward.

Markets are the rational theory of reward

Setup. There is a vector space $\mathbb{Q}$ in which quantity vectors live (its dimensionality is the number of unique goods). There is a directed graph, where every node i is a "factory" labelled with production function $f_i : \mathbb{Q} \to \mathbb{Q}$ and every edge ij is labelled by a quantity vector $q_{ij} \in \mathbb{Q}$, subject to physical constraints $f_i(x_i) = y_i$ where $x_i = \sum_k q_{ki}$ and $y_i = \sum_j q_{ij}$. Some nodes, called "consumers", have a utility function $U_i(x_i, y_i)$; the global welfare objective is $W = \sum_i U_i(x_i, y_i)$.

This is a constrained optimization problem; to solve it we must parameterize quantity flows with routing shares $q_{i\to j} = s_{ij} \odot y_i$. Then we have the forward pass

$$x_i = r_i + \sum\nolimits_h s_{hi} \odot y_h, \qquad y_i = f_i(x_i)$$

and backward pass

$$a_i = \tfrac{\partial W}{\partial x_i} = \nabla_{x_i} U_i + J_{f_i}(x_i)^\top b_i \tag{1.2}$$

$$b_i = \tfrac{\partial W}{\partial y_i} = \nabla_{y_i} U_i + \sum\nolimits_j s_{ij} \odot a_j \tag{1.3}$$

$$c_{ij} = \tfrac{\partial W}{\partial s_{ij}} = y_i \odot a_j \tag{1.4}$$

In fact our market is equivalent to an MDP (with matrix-valued discount factor $J_{f_i}^\top$), and this is a Bellman recursion:

RL	market
state	"one unit of good g sits at node i"
action	"ship it to j"
policy $\pi(j \mid i)$	routing share $s_{ij,g}$
reward	marginal utility ∇U collected along the way
action value $Q(i,j)$	destination price $a_{j,g}$
state value $V(i)$	average received price $\bar{a}_{i,g}$
advantage $A(i,j)$	price premium $a_{j,g} - \bar{a}_{i,g}$

Now observe: the market "learning" involves two processes simultaneously:

1. **Agents learn the shares** s_{ij}, e.g. through continuous-time Bayesian inference with revenue c as the evidence.
2. **Price recursion**, i.e. equations (1.2)–(1.4), which follow a Bellman/backprop-like rule.

In other words: **the economy is exactly the "super-agent" that generates reward to its sub-agents, and price recursion is the "rational theory of reward".**